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SUBJECT NAME: MACHINE LEARNING

EXPERIMENT-01

IMPLEMENT AND DEMONSTRATE THE FIND-S ALGORITHM FOR FINDING THE MOST SPECIFIC HYPOTHESIS BASED ON A GIVEN SET OF TRAINING DATA SAMPLES.

1: FIND-S ALGORITHM FOR ENJOY SPORT DATASET

AIM

To implement the FIND-S algorithm and obtain the most specific hypothesis that is consistent with the positive training examples in the Enjoy Sport dataset.

APPARATUS / SOFTWARE REQUIRED

Computer or laptop, Python 3.x interpreter (IDLE, Jupyter Notebook, or any Python IDE), and the given training dataset.

PROCEDURE

1)Enter the training examples shown in the dataset.

2)Initialize the hypothesis as None.

3)Process each example and update the hypothesis only for target value Yes.

4)For each positive example after the first, replace any mismatching attribute with ?.

5)Display the final hypothesis and interpret its attributes.

DATASET DESCRIPTION

The dataset contains six descriptive attributes—Sky, AirTemp, Humidity, Wind, Water, and Forecast—and one target attribute, Enjoy Sport. The target value is Yes for positive examples and No for negative examples. FIND-S begins with the first positive example and generalizes an attribute to ? whenever later positive examples have a different value for that attribute. Negative examples are ignored.

Sky	AirTemp	Humidity	Wind	Water	Forecast	Enjoy Sport
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Change	Yes

PROGRAM

```
# FIND-S: Enjoy Sport
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]
hypothesis = None
for example in data:
    if example[-1] == 'Yes':
        attributes = example[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = '?'
print('Final hypothesis:', hypothesis)
```

INPUT

The input is the training dataset shown under DATASET DESCRIPTION. Each row contains the attribute values followed by the target label (Yes or No).

OUTPUT

Final hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

RESULT

The FIND-S algorithm was implemented successfully. The final hypothesis is <Sunny, Warm, ?, Strong, ?, ?>. For the positive examples, Sky is Sunny, AirTemp is Warm, and Wind is Strong; Humidity, Water, and Forecast may take any value.

2: FIND-S ALGORITHM FOR PLAY CRICKET PREDICTION

AIM

To implement the FIND-S algorithm and find the most specific hypothesis for predicting whether a person will play cricket.

APPARATUS / SOFTWARE REQUIRED

Computer or laptop, Python 3.x interpreter (IDLE, Jupyter Notebook, or any Python IDE), and the given training dataset.

PROCEDURE

1)Enter the training examples shown in the dataset.

2) Initialize the hypothesis as None.

3)Process each example and update the hypothesis only for target value Yes.

4) For each positive example after the first, replace any mismatching attribute with ?.

5)Display the final hypothesis and interpret its attributes.

DATASET DESCRIPTION

The dataset uses Weather, Temperature, Humidity, and Wind as input attributes. Play Cricket is the target attribute. Examples labeled Yes are used to update the hypothesis; the No example is ignored by FIND-S.

Weather	Temperature	Humidity	Wind	Play Cricket
Sunny	Hot	High	Weak	Yes
Sunny	Hot	High	Strong	Yes
Rainy	Cool	Normal	Weak	No
Sunny	Mild	High	Weak	Yes

PROGRAM

```
# FIND-S: Play Cricket
data = [
    ['Sunny', 'Hot', 'High', 'Weak', 'Yes'],
    ['Sunny', 'Hot', 'High', 'Strong', 'Yes'],
```

```
    ['Rainy', 'Cool', 'Normal', 'Weak', 'No'],
    ['Sunny', 'Mild', 'High', 'Weak', 'Yes']
]
hypothesis = None
for example in data:
    if example[-1] == 'Yes':
        attributes = example[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = '?'
print('Final hypothesis:', hypothesis)
```

INPUT

The input is the training dataset shown under DATASET DESCRIPTION. Each row contains the attribute values followed by the target label (Yes or No).

OUTPUT

Final hypothesis: ['Sunny', '?', 'High', '?']

RESULT

The FIND-S algorithm was implemented successfully. The final hypothesis is <Sunny, ?, High, ?>. It matches the positive examples with Sunny weather and High humidity, while Temperature and Wind can take any value.

3: FIND-S ALGORITHM FOR STUDENT PASS PREDICTION

AIM

To implement the FIND-S algorithm and find the most specific hypothesis for predicting whether a student will pass an examination.

APPARATUS / SOFTWARE REQUIRED

Computer or laptop, Python 3.x interpreter (IDLE, Jupyter Notebook, or any Python IDE), and the given training dataset.

PROCEDURE

1)Enter the training examples shown in the dataset.

2)Initialize the hypothesis as None.

3)Process each example and update the hypothesis only for target value Yes.

4)For each positive example after the first, replace any mismatching attribute with ?.

5)Display the final hypothesis and interpret its attributes.

DATASET DESCRIPTION

The dataset contains Study Hours, Attendance, Assignment, and Sleep as input attributes. Pass is the target attribute. Positive examples are labeled Yes. The algorithm generalizes only attributes that differ among positive examples.

Study Hours	Attendance	Assignment	Sleep	Pass
High	Good	Complete	Good	Yes
High	Good	Complete	Poor	Yes
Low	Poor	Incomplete	Good	No
High	Good	Incomplete	Good	Yes

PROGRAM

```
# FIND-S: Student Pass Prediction
data = [
    ['High', 'Good', 'Complete', 'Good', 'Yes'],
    ['High', 'Good', 'Complete', 'Poor', 'Yes'],
```

```
    ['Low', 'Poor', 'Incomplete', 'Good', 'No'],
    ['High', 'Good', 'Incomplete', 'Good', 'Yes']
]
hypothesis = None
for example in data:
    if example[-1] == 'Yes':
        attributes = example[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = '?'
print('Final hypothesis:', hypothesis)
```

INPUT

The input is the training dataset shown under DATASET DESCRIPTION. Each row contains the attribute values followed by the target label (Yes or No).

OUTPUT

Final hypothesis: ['High', 'Good', '?', '?']

RESULT

The FIND-S algorithm was implemented successfully. The final hypothesis is <High, Good, ?, ?>. The positive examples share High study hours and Good attendance; Assignment and Sleep are generalized to any value.

4: FIND-S ALGORITHM FOR LAPTOP PURCHASE PREDICTION

AIM

To implement the FIND-S algorithm and identify the most specific hypothesis for predicting whether a customer will purchase a laptop.

APPARATUS / SOFTWARE REQUIRED

Computer or laptop, Python 3.x interpreter (IDLE, Jupyter Notebook, or any Python IDE), and the given training dataset.

PROCEDURE

1)Enter the training examples shown in the dataset.

2)Initialize the hypothesis as None.

3)Process each example and update the hypothesis only for target value Yes.

4)For each positive example after the first, replace any mismatching attribute with ?.

5)Display the final hypothesis and interpret its attributes.

DATASET DESCRIPTION

The input attributes are Price, Brand, RAM, and Storage. Purchase is the target attribute. FIND-S starts from the first Yes example and replaces differing attributes with ? when another positive example is processed. The negative example does not change the hypothesis.

Price	Brand	RAM	Storage	Purchase
Low	A	8GB	SSD	Yes
Low	A	16GB	SSD	Yes
High	B	8GB	HDD	No
Low	A	8GB	HDD	Yes

PROGRAM

```
# FIND-S: Laptop Purchase Prediction
data = [
    ['Low', 'A', '8GB', 'SSD', 'Yes'],
    ['Low', 'A', '16GB', 'SSD', 'Yes'],
```

```
    ['High', 'B', '8GB', 'HDD', 'No'],
    ['Low', 'A', '8GB', 'HDD', 'Yes']
]
hypothesis = None
for example in data:
    if example[-1] == 'Yes':
        attributes = example[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = '?'
print('Final hypothesis:', hypothesis)
```

INPUT

The input is the training dataset shown under DATASET DESCRIPTION. Each row contains the attribute values followed by the target label (Yes or No).

OUTPUT

Final hypothesis: ['Low', 'A', '?', '?']

RESULT

The FIND-S algorithm was implemented successfully. The final hypothesis is <Low, A, ?, ?>. In the positive training samples, Price is Low and Brand is A, while RAM and Storage may take any value.

5: FIND-S ALGORITHM FOR OUTDOOR ACTIVITY PREDICTION

AIM

To implement the FIND-S algorithm and find the most specific hypothesis for predicting whether an outdoor activity will take place.

APPARATUS / SOFTWARE REQUIRED

Computer or laptop, Python 3.x interpreter (IDLE, Jupyter Notebook, or any Python IDE), and the given training dataset.

PROCEDURE

1)Enter the training examples shown in the dataset.

2)Initialize the hypothesis as None.

3)Process each example and update the hypothesis only for target value Yes.

4)For each positive example after the first, replace any mismatching attribute with ?.

5)Display the final hypothesis and interpret its attributes.

DATASET DESCRIPTION

The dataset contains Weather, Temperature, Wind, and Forecast as input attributes. Activity is the target attribute. The algorithm considers the three positive examples and ignores the negative example while forming the final hypothesis.

Weather	Temperature	Wind	Forecast	Activity
Sunny	Warm	Weak	Clear	Yes
Sunny	Warm	Strong	Clear	Yes
Rainy	Cold	Strong	Cloudy	No
Sunny	Cold	Weak	Clear	Yes

PROGRAM

```
# FIND-S: Outdoor Activity Prediction
data = [
    ['Sunny', 'Warm', 'Weak', 'Clear', 'Yes'],
    ['Sunny', 'Warm', 'Strong', 'Clear', 'Yes'],
```

```
    ['Rainy', 'Cold', 'Strong', 'Cloudy', 'No'],
    ['Sunny', 'Cold', 'Weak', 'Clear', 'Yes']
]
hypothesis = None
for example in data:
    if example[-1] == 'Yes':
        attributes = example[:-1]
        if hypothesis is None:
            hypothesis = attributes.copy()
        else:
            for i in range(len(hypothesis)):
                if hypothesis[i] != attributes[i]:
                    hypothesis[i] = '?'
print('Final hypothesis:', hypothesis)
```

INPUT

The input is the training dataset shown under DATASET DESCRIPTION. Each row contains the attribute values followed by the target label (Yes or No).

OUTPUT

Final hypothesis: ['Sunny', '?', '?', 'Clear']

RESULT

The FIND-S algorithm was implemented successfully. The final hypothesis is <Sunny, ?, ?, Clear>. The positive examples share Sunny weather and a Clear forecast; Temperature and Wind may take any value.